Understanding the Impact of Turbulence on Floating Lidar Measurements

Andreu Salcedo-Bosch, Francesc Rocadenbosch, *Senior Member, IEEE*, Alfredo Peña, Jakob Mann, and Simone Lolli

***Abstract*— Retrieving accurate turbulence intensity (TI) from motion-corrupted floating lidar measurements is one of the main challenges of the offshore wind energy industry today. Toward overcoming this challenge, this work describes a method to retrieve the motion-corrected TI from floating lidar measurements. The method relies on an atmospheric turbulence model combined with lidar-based flow characteristics estimates. The method simulates floating-lidar and anemometer-like measurements via synthetically generated wind fields in turbulence boxes. We highlight the importance of accurate modeling of turbulence-box parameters to virtually eliminate motion-induced error. The proposed method was successfully tested over a 3-month campaign at IJmuiden by comparing floating-lidar turbulence measurements against those from measurements from a reference meteorological mast.**

***Index Terms*— Anemometer, lidar, Mann model, meteorological mast, ocean, offshore wind energy, remote sensing, simulator, surface layer, turbulence, wind energy.**

I. Introduction

IN THE last decades, the wind energy industry has made important investments in the offshore wind resource due to the higher and more homogeneous winds that can be found over the ocean [1], [2]. Similar to onshore wind, one of the main concerns of the industry is the accurate measurement of the wind resource. Traditionally, this has been carried out by heavily instrumented meteorological masts (metmasts) planted on the seabed [3]. As wind farms are being deployed farther from shore and into deeper waters [4], metmasts are not a feasible solution, and remote sensing techniques such as satellite [5], [6], radar [7], sodar [8], or Doppler wind lidar (DWL) [9] are used as alternatives.

Today, DWLs deployed on floating platforms or buoys, namely floating DWLs (FDWL), are the de facto instrument for the evaluation of offshore wind resources due to their ability to detect the wind profile accurately, flexibly, and cost-effectively [10], [11], [12], [13]. Compared with anemometers deployed in offshore metmasts, FDWLs provide nearly identical mean horizontal wind speed (HWS) and wind direction (WD) measurements (coefficient of determination $R^2 > 0.98$) at the 10-min level [14], [15], [16], [17].

Recent studies have demonstrated that FDWLs can independently retrieve surface-layer parameters without additional instrumentation [12], [18], [19]. For example, friction velocity (u_*) and Obukhov length (L) can be derived using the 2P-algorithm of Araújo da Silva et al. [12], which fits the Monin–Obukhov wind profile model to observations of the vertical profile of HWS [12]. A review of the 2P algorithm is given in Appendix A.

FDWL measurements of the wind vector are affected by two error sources: motion-induced error and the inherent temporal and spatial averaging of the DWL. These errors are averaged out on the 10-min mean wind vector [12], [20], [21], but arise at higher statistical moments of wind, such as its variance, which is related to a parameter used to characterized turbulence, the so-called turbulence intensity (TI). This is defined as the ratio of the standard deviation of the HWS to its mean.

TI is used by the wind energy industry for assessing the conditions for the deployment and operation of wind farms. Erroneous TI measurements in the wind siting phase can lead to over/underdesign and malfunctioning of wind turbines, increasing the deployment and maintenance costs of wind farm projects [16], [22]. Therefore, estimation and compensation of the measurement error of TI has been a hot topic in recent years [16], [20], [21], [23], [24], [25].

Regarding motion-induced error, FDWLs experience a 6-degree-of-freedom (DoF) motion due to ocean waves motion [20], [26], [27], [28], inducing errors in instantaneous wind vector measurements [29], [30]. This results in a variance increase at the 10-min level, which is measured by an FDWL as an apparent turbulence increase [20], [21], [28].

Compensation of motion-induced TI in continuous-wave (CW) DWLs has been tackled by means of multiple methods. Tiana-Alsina et al. [30] considered mechanical motion

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compensation by deploying the DWL on a cardanic frame. This method was able to compensate for rotational motion, but it increased the hardware cost of the device and could not compensate for translational motion. Gutiérrez-Antuñano et al. [14] proposed a simple window-averaging technique by which the HWS temporal series was smoothed as a function of the FDWL tilting period. Unfortunately, this technique requires that the FDWL wind vector be retrieved at a faster rate than the wave-induced motion frequency, which is not always the scenario encountered. Gutiérrez-Antuñano et al. [15] also presented a statistical error correction approach based on an FDWL motion simulator. The simulator considered basic motion parameters (tilting period and amplitude) and simulated the motion-induced TI measurement error under a homogeneous wind flow. This method showed an acceptable correction, but the results depended on the atmospheric scenario encountered. Kelberlau et al. [16] presented a correction algorithm that compensates for the motion effects on the radial velocity measurements by the FDWL. However, this method requires access to internal radial velocity measurements, which is usually undisclosed information by the lidar manufacturer. Salcedo-Bosch et al. [23], [24] presented an adaptive unscented Kalman filter capable of compensating for motion-induced error on the run without having access to internal radial velocity measurements. By modeling the wind as an autoregressive process, the filter was able to deconvolve the "true" wind vector, thus allowing the retrieval of highly accurate TI measurements with reference to a fixed DWL ($\rho^2 = 0.96$). Nevertheless, it was unable to compensate for the DWL's inherent spatial and temporal averaging of the wind vector.

Regarding the inherent averaging of CW DWLs, they retrieve the wind vector from a series of line-of-sight (LoS) measurements following a conical scanning pattern [9]. Therefore, they measure a temporally and spatially averaged version of the wind vector [9], [31], [32], inherently averaging wind gusts or eddies with sizes that fit within the DWL scanning cone. This results in a miss-estimation in the variance of the velocity measurements and, therefore, corrupts TI FDWL measurement values [33], [34]. Thus, CW FDWLs suffer from both the inherent averaging effect and the motion-induced error as compared to fixed anemometers.

Recently, Peña et al. [35] went a step further and aimed to compensate for the motion, the inherent averaging effect, and cross-contamination on FDWLs to obtain anemometer-like measurements of TI. The method involved simulating FDWL measurements in synthetic turbulent wind fields and comparing the FDWL-retrieved TI to the "true" TI. Promising results were achieved, significantly reducing TI measurement bias in most scenarios. However, undercorrection was found at high altitudes and low HWSs.

The turbulent wind fields used for the simulations in [35] were generated via stochastic wind field realizations based on the turbulence spectral tensor model of Mann [36], [37]. The Mann spectral tensor characterizes the energy, length, and degree of anisotropy of turbulence using three parameters: the dissipation rate of turbulence ($\alpha\epsilon^{2/3}$), the turbulence length scale (L_{MM}), and the eddy life-time parameter (Γ). Mann parameters are typically obtained by fitting the Mann spectral tensor to the measured velocity spectra [38]. Due to the low temporal resolution of FDWL wind measurements (lower than the main frequency components of turbulence), Mann parameters could not be derived and were considered invariant, leading to turbulence boxes that are stochastically generated but with nearly equal turbulence statistics. Moreover, turbulence boxes usually do not account for variations in turbulence characteristics with height [39].

In this study, we aim at improving the correction method proposed by Peña et al. [35] by incorporating synthetic wind fields that account for variations in the turbulence spectral characteristics. The aim is to account for changes in turbulence amplitude and length scale, as described by the Mann spectral tensor, to better represent real atmospheric conditions. Toward this purpose, we review empirical expressions that relate the turbulence amplitude and length scale to surface-layer parameters, that is, friction velocity (u_*) and the vertical wind local gradient (dU/dz), which can be inferred from FDWL-measured HWS profiles. Both the original and improved correction methods are evaluated over experimental data from the IJmuiden campaign, where FDWL measurements were compared against a reference metmast, and the results are analyzed across different atmospheric and motion conditions.

This article is structured as follows. Section II introduces the IJmuiden campaign and summarizes the instrumentation used. Section III reviews the correction method by Peña et al. [35] and presents the improved method. Section IV compares and discusses the performance of Peña's et al. and the improved method as a function of different atmospheric and motion scenarios. Section V gives concluding remarks.

II. Materials

The EOLOS[1] FDWL validation campaign was carried out at the IJmuiden meteorological site, which is located in the North Sea (52.848 N, 3.436 E), 85 km from the Netherlands, and characterized by shallow waters about 28 m deep. During 81 days, from March 29 to June 17, 2015, the FDWL was located 200 m away from the IJmuiden metmast and measured WD and HWS at 25, 38, 56, and 83 m. These lidar measurement heights enabled virtually height-coincident measurements with those from the mast-anemometers installed at 27, 58.5, and 85 m above the lowest astronomical tide (LAT) [12].

A. Floating DWL

The floating-system EOLOS was a lidar buoy that hosted a ZephIR[1] 300 DWL manufactured by ZX Lidars Ltd. and a 3DM-GX2-45 inertial measurement unit (IMU) to record the motion of the lidar buoy. Its 3.77-m width, 3-ton weighted structure was covered by an aluminum-reinforced fiber–glass material to protect internal hardware [15]. It had a four-floater structure design, with one of the 4 corners having a mounted tail which acted as a "stern" to align it with the WD.

The ZephIR[1] 300 is a CW DWL measuring the vertical profile of the wind speed at user-defined heights of 10–200 m

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[40]. For a single 3D-wind-vector retrieval at a measurement height h, ZephIR[1] 300 measures the Doppler radial velocities along 50 LoSs following a scanning cone of $\theta_0 = 30°$ aperture from the zenith. The conical scanning pattern is created using a prism rotating at a constant uniform circular motion of 1 revolution per second that deflects the light beam emitted by the DWL. With a constant homogeneous wind, the measured LoSs take the shape of a cosine as a function of the azimuth scanning angle ϕ. The DWL can retrieve the 3-D wind vector by fitting a cosine function to the LoSs through the so-called velocity azimuth display (VAD) algorithm [23]. The ZephIR[1] 300 operates by sequentially scanning each configured measurement height [41]. Consequently, under the IJmuiden campaign setup, which involved four measurement heights, the temporal resolution at any given height is approximately 4 s.

The 3DM-GX2-45 IMU combines a high-precision GPS antenna, a gyro, and an accelerometer to provide high-resolution measurements of the 6-DoF attitude of the EOLOS FDWL buoy. This IMU measures the buoy's tilt and velocity at a resolution of 10 Hz in the north-east-down global frame of reference.

To ensure time synchronization between all sensors of the FDWL, data acquisition and control are centralized using a CR3000 Micrologger (Campbell Scientific, Inc.) [42], suitable for long-term environmental monitoring missions. The data could be accessed through an Iridium satellite or via a Wi-Fi link within a 100-m range.

B. IJmuiden Metmast

A detailed description of IJmuiden instrumentation can be found in [43]. For brevity, we summarize only the sensors used in the present work. The 92-m mast structure was made up of three booms oriented to 46.5°, 166.5°, and 286.5°, clockwise from the cardinal north. Point-wise high-resolution wind speed observations were gathered by six Thies[1] first-class advanced cup anemometers, which were installed at 27 and 58.5 m in height (one on each boom), and three Metek[1] USA-1 sonic anemometers at 85 m. Nine Thies[1] first-class wind-vanes recorded the WD at 26.2, 57.7, and 87 m (one on each boom).

III. Methods

A. Basic Definitions and Foundations

1) Wind Parameters: The instantaneous wind vector $\vec{u}$ is defined as the three-component vector

$$\vec{u} = \left[u_x, u_y, u_z\right] \tag{1}$$

where u_x, u_y, and u_z are the wind vector components in the x-, y-, and z-directions, respectively. The instantaneous wind vector can be defined as the addition of the 10-min mean wind vector ($\overline{\vec{u}}$) and the instantaneous fluctuations around the mean ($\tilde{\vec{u}}$). Formally

$$\vec{u}(\boldsymbol{x}) = \left[\overline{u}_x + \tilde{u}_x\left(\vec{x}\right), \overline{u}_y + \tilde{u}_y\left(\vec{x}\right), \overline{u}_z + \tilde{u}_z\left(\vec{x}\right)\right] \tag{2}$$

where $\overline{u}_i + \tilde{u}_i(\vec{x})$ is the instantaneous wind vector component $\vec{u}_i$, and $\vec{x} = (x, y, z)$ is the position vector.

The HWS, denoted by U, can be obtained as the quadratic sum of the x and y wind vector components as

$$U = \sqrt{{u_x}^2 + {u_y}^2}. \tag{3}$$

The 10-min TI is defined as

$$\mathrm{TI} = \frac{\sigma_U}{\overline{U}} \tag{4}$$

where σ_U and $\overline{U}$ are the 10-min standard deviation and mean HWS, respectively.

2) Surface-Layer Parameters: The surface layer is the bottom region of the boundary layer, in which turbulent fluxes and stress show variations of less than 10% [44]. Monin–Obukhov similarity theory (MOST, [45]) provides a framework for understanding how vertical gradients of heat and momentum behave in the surface layer. According to MOST, under a neutral, stationary, and homogeneous flow [46]

$$\frac{dU}{dz} = \frac{u_*}{\kappa z} \tag{5}$$

where u_* is the friction velocity and $\kappa \simeq 0.4$ is the von Kármán constant.

The friction velocity (u_*) is indicative of the vertical flux of the horizontal air momentum, defined as

$$u_* = \left[\overline{\tilde{u}_x\tilde{u}_z}^2 + \overline{\tilde{u}_y\tilde{u}_z}^2\right]^{1/2}. \tag{6}$$

According to MOST, in neutral atmospheres, where turbulence is purely mechanically generated, friction velocity relates to the variance of the HWS. From extensive experimental measurement campaigns in flat terrain [47], [48]

$$\sigma_U \simeq 2.93u_*. \tag{7}$$

The surface layer local gradient dU/dz and the friction velocity u_* cannot be directly measured by the EOLOS[1] FDWL used in IJmuiden. However, they can be inferred from the wind profiles measured by the FDWL using the 2P algorithm and the Högström polynomial model, defined in Appendix A.

3) Mann's Turbulence Model: The Mann turbulence model can be used to generate 2-D and 3-D random turbulent wind fields with predefined turbulence characteristics [36], also known as turbulence boxes in a domain of dimensions $L_x \times L_y \times L_z$, consisting of $n_x \times n_y \times n_z$ grid points. These wind fields are, therefore, characterized by the three Mann parameters: 1) the *dissipation rate of turbulence* ($\alpha\epsilon^{2/3}$), 2) the *turbulence length scale* (L_{MM}), and 3) the *eddy lifetime parameter* (Γ). Although being developed for neutral stratification, the Mann model has been shown to represent experimental wind turbulence across most atmospheric stability regimes by adjusting its parameters [38], [39]. It performs worst under very convective conditions, and it is important to note that large-scale turbulence structures present in the marine boundary layer are not captured by the model [49], [50]. Nevertheless, low-frequency turbulence can be reliably measured by DWLs [35] and is, therefore, not the focus of this study. Detailed information on the Mann model and turbulence energy spectra can be found in [36] and [37].

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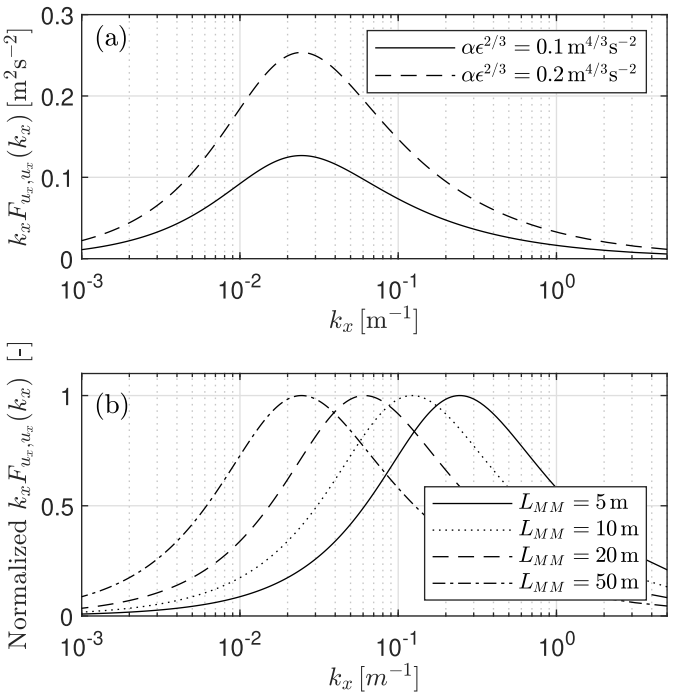


Fig. 1. Comparison of wind turbulence spectra for different Mann's model parameters. $F_{u_x,u_x}(k_x)$ denotes the turbulent spectrum of wind component u_x. (a) Scaling effect due to model parameter $\alpha\epsilon^{2/3}$. (b) Peak spectrum shift due to model parameter L_{MM}.

As an example, Fig. 1 illustrates the role of the turbulence length scale and dissipation rate parameters in determining the turbulent spectra of the wind component u_x [denoted $F_{u_x,u_x}(k_x)$] scaled as function of the wavenumber $k_x = 2\pi x/U$.

In panel (a), we see how $\alpha\epsilon^{2/3}$ scales the turbulence spectra, acting as a scaling factor of the turbulent wind fluctuations. At this point, it is important to note that $\alpha\epsilon^{2/3}$ is proportional to the variance of the wind components ($\sigma^2_{u_x}$, $\sigma^2_{u_y}$, and $\sigma^2_{u_z}$), which are the integrals of the corresponding turbulence spectra, that is, $\sigma^2_{u_x} = \int_{-\infty}^{\infty} F_{u_x,u_x}(k_x)\mathrm{d}k_x$, $\sigma^2_{u_y} = \int_{-\infty}^{\infty} F_{u_y,u_y}(k_y)\mathrm{d}k_y$, and $\sigma^2_{u_z} = \int_{-\infty}^{\infty} F_{u_z,u_z}(k_z)\mathrm{d}k_z$. Therefore, we use the HWS standard deviation instead of the rate of dissipation to describe turbulence.

The turbulence length scale (L_{MM}) represents the size of the largest energy-containing eddies. Panel (b) illustrates how L_{MM} sets the location of the peak of the turbulence spectrum. The turbulence length scale is proportional to the mixing length, which, in terms of friction velocity and wind shear gradient, can be expressed as

$$L_{\mathrm{MM}} \simeq k\frac{u_*}{dU/dz} \tag{8}$$

being the proportionality constant $k \approx 2.3$ in flat terrain [38], [51].

Finally, the eddy life-time parameter (Γ) defines the turbulence anisotropy, characterizing how eddies deform and evolve over time and space. Γ affects the distribution of turbulent energy between cross-wind and along-WDs, rather than directly impacting the overall turbulent energy magnitude.

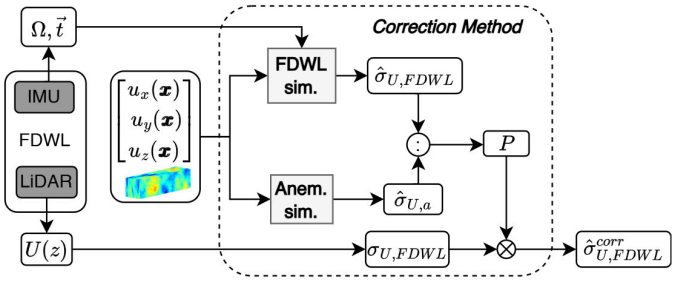


Fig. 2. Computational diagram describing the processing chain to correct FDWL measurements using turbulence boxes. Operational blocks ":" and "×" refer to division and multiplication operations, respectively.

B. Motion Compensation of Floating DWL TI Measurements: A Review

Peña et al. [35] presented a first approach for compensating FDWL TI measurement errors using Mann-based turbulence boxes. The procedure is depicted in Fig. 2. For each FDWL 10-min measurement period, they used a Mann turbulence box to simulate the radial velocities scanned by a CW FDWL, reconstructed the velocity components with the VAD algorithm, and computed the HWS standard deviation ($\hat{\sigma}_{U,\mathrm{FDWL}}$) subjected to the rotational and translational motions (Ω and $\vec{t}$, respectively) that were recorded by the buoy's IMU. They also simulated the HWS standard deviation measurement of an idealized sonic anemometer ($\hat{\sigma}_{U,a}$) in the same turbulence box. Details on the FDWL and sonic anemometer wind measurement simulations in turbulence boxes are given in Appendices B2 and B3.

To minimize random errors, the process was repeated using 100 different turbulence boxes, each generated with a distinct random seed while maintaining the same Mann parameters, $\alpha\epsilon^{2/3} = 0.05\ \mathrm{m}^{4/3}\mathrm{s}^{-2}$, $L_{\mathrm{MM}} = 61$ m, and $\Gamma = 3.2$, and hence using turbulence boxes with constant turbulence characteristics. Although promising results were obtained, undercompensation was observed at high altitudes and low wind speeds. They argued that this was partly due to the invariant characteristics of the turbulence boxes used.

We conduct a parametric study to analyze how DWL and FDWL measurements are affected by the different spectral characteristics of the turbulence boxes, namely, we study results for a range of σ_U, L_{MM}, and Γ values. The simulation parameters used are summarized in Table I. The Mann parameters ($\alpha\epsilon^{2/3}$, L_{MM}, and Γ) ranges are defined according to the domain values obtained from sonic anemometer measurements at 10-, 20-, 40-, 60-, 80-, and 100-m heights at the National Test Station for Wind Turbines at Høvsøre, Denmark, during 11 months [38], resulting in a total of 29 400 boxes. Note that the Mann turbulence boxes show, on average, the same relation between u_* and σ_U as that described in 7.

We compute the HWS standard deviation measured by a sonic anemometer ($\sigma_{U,a}$), a fixed DWL ($\sigma_{U,\mathrm{DWL}}$), and an FDWL ($\sigma_{U,\mathrm{FDWL}}$) for each turbulence box. We simulate the measurements performed at 83-m height, which is the highest measurement height of the FDWL in IJmuiden, corresponding to a scanning cone diameter of 95.8 m. The parametric study results are given in Section IV-A.

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TABLE I
MANN'S MODEL PARAMETERS USED IN THE TURBULENCE-BOX PARAMETRIC STUDY

Parameter	Units	Range Values
$\alpha\epsilon^{2/3}$	$m^{4/3}\ s^{-2}$	0.05
L_{MM}	m	From 0 to 70 in steps of 2
Γ	-	From 1.4 to 4.4 in steps of 0.2
σ_U	$m\ s^{-1}$	From 0 to 2.5 in steps of 0.3
L_x, L_y, L_z	m	18000, 128, 128 m
n_x, n_y, n_z	# of points	4096, 32, 32
Seeds	-	From -1 to -21 in steps of -3

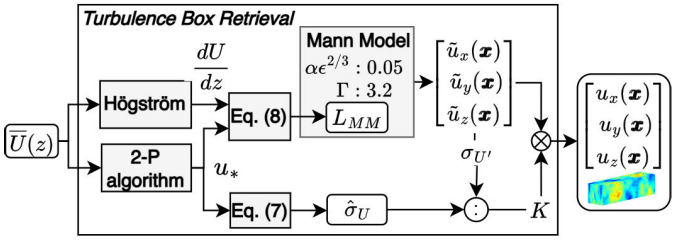


Fig. 3. Computational diagram describing the processing chain to retrieve the adequate turbulence boxes generated by the Mann model. Operational blocks ":" and "×" refer to division and multiplication operations, respectively.

C. Motion Compensation of Floating DWL TI Measurements: Enhanced Method

Based on the approach of Peña et al. [35], hereafter referred to as the *basic method*, we propose an *enhanced* FDWL correction method. This enhanced method also uses turbulence boxes but with characteristics close to the real turbulent conditions experienced by the FDWLs.

The parameters of the Mann model can be indirectly estimated from surface-layer parameters (e.g., u_* and dU/dz), which can be computed by FDWLs via the 2P algorithm (see Appendix A).

The procedure for retrieving the turbulence boxes is synthesized in Fig. 3. Note that the mean vertical profile of HWS ($\overline{U(z)}$) is the only input required, which can be accurately measured by FDWLs [12].

The procedure involves four main steps.

1) The friction velocity (u_*) and the wind shear gradient (dU/dz) are retrieved from the vertical profile measured by the FDWL mean HWS ($\overline{U}(z)$) through the 2P algorithm and Högström polynomial model (see Appendix A), respectively.
2) The length scale (L_{MM}) and the expected HWS standard deviation ($\hat{\sigma}_U$) are estimated from u_* and dU/dz through 7 and 8.
3) The Mann model is used to simulate turbulence boxes $[\tilde{u}_x(\boldsymbol{x}), \tilde{u}_y(\boldsymbol{x}), \tilde{u}_z(\boldsymbol{x})]$ with $\alpha\epsilon^{2/3} = 0.05$, $\Gamma = 3.2$, and L_{MM} from step 2 as input parameters for the model. Such boxes are characterized by $\sigma_{U'}$.
4) The boxes wind vectors are equally scaled to have the expected HWS standard deviation ($\hat{\sigma}_U$) through factor $K = \hat{\sigma}_U/\sigma_{U'}$.

Once the scaled turbulence boxes are obtained, they are used to correct the FDWL measurements similarly to the basic method (see Section III-B and Fig. 2).

D. Data Used and Outlier Filtering Criteria

The dataset used for the model validation comprised data from March 29, 2015, to June 17, 2015. Specifically: 1) wind measurements by the FDWL at 25, 56, and 83 m above LAT, 2) wind measurements by the cup and sonic anemometers at 27, 58.5, and 85 m above LAT, 3) FDWL 6-DoF motion measurements by the IMU, and 4) FDWL data quality parameters. Data were filtered according to the following outlier rejection criteria [12]: data records with FDWL measured HWS outside the 1–80-m s^{-1} range, rain-flagged, data with spatial variation greater than 0.2 (associated with erroneous HWS measurements [12]), backscatter coefficients lower than 0.02, or invalid or NaN IMU measurements were removed.

To quantitatively assess and compare the instruments, the coefficient of determination (R^2), root-mean-square error (RMSE), and mean deviation (MD) indicators were used.

E. Hardware Specifications

Turbulence box simulations and FDWL measurement corrections were performed using MATLAB[2] R2020a on a desktop PC with the following specifications: Intel[2] Core[1] i9-9900K CPU @ 3.60 GHz (8 cores), 32 GB of RAM, and a 1 TB hard drive. With the system fully dedicated and leveraging parallelization, approximately 140 GB of storage and 180 min were required to generate the turbulence boxes, while processing the entire IJmuiden campaign took approximately 900 min (15 h; see Section IV-C).

IV. RESULTS AND DISCUSSION

A. Parametric Study Results

This section presents and discusses the results of the parametric study in Section III-B. Fig. 4 shows the performance of each instrument [DWL in panel (a) and FDWL in panel (b)], which is expressed in terms of the HWS standard deviation measured normalized to the "true" atmospheric HWS standard deviation, $\sigma_{U,a}$. Our findings are as follows.

First, we find that Γ has minimal impact on the ratios of HWS standard deviation explored. Second, Fig. 4(a) shows that the length scale, L_{MM}, has a strong impact on the DWL turbulence measurements, as expected. Under low L_{MM} values, the DWL strongly underestimates the reconstructed wind speed variance, with values up to 50%. This is inherent to the DWL scans, which average out small turbulent eddies [52]. By using the fixed length scale considered by Peña et al. [35] ($L_{\mathrm{MM}} = 61$, m, blue dashed line), the averaging of small turbulent eddies by DWLs is not that strong, which leads to undercompensation particularly at the highest measurement levels, where the DWL scanning cone is larger, and where the averaging along the beam is larger for this type of lidar. Third, in Fig. 4(b), we analyze the impact on turbulence ratios under rotational and translational motions, separately. Each component of the motion is simulated as a simple harmonic motion with an oscillatory frequency of 0.3 s, a typical value observed by offshore FDWL deployed [53]. For simplicity, 1-DoF surge and rotational motions are considered.

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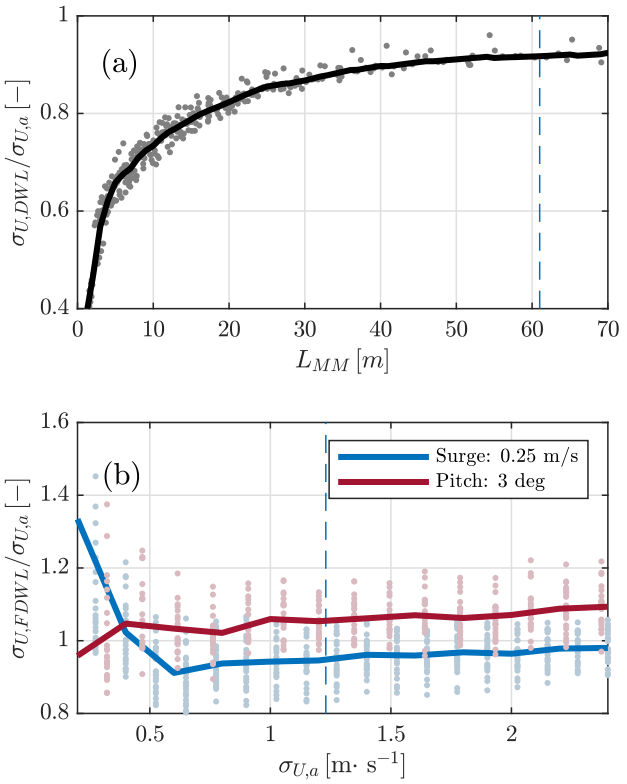


Fig. 4. DWL and FDWL HWS standard deviation, $\sigma_{U,DWL}$ and $\sigma_{U,\mathrm{FDWL}}$, normalized to reference HWS standard deviation ($\sigma_{U,a}$) (as measured by sonic anemometer) as a function of different turbulent parameters. (a) DWL-to-anemometer ratio as a function of turbulence-length scale L_{MM}. (b) FDWL-to-anemometer ratio as a function of reference HWS standard deviation $\sigma_{U,a}$. (Blue line) FDWL is subject to translational motion only. (Red line) Pitch rotational motion only. Simulation parameters are summarized in Table I. (Dots) Results from each random simulation seed. (Thick lines) Interpolated average of all random seeds. (Blue dashed) Fixed parameters: $L = 61$ m and $\sigma_U \simeq 1.34\, m^{4/3} s^{-2}$ used in [35].

Results in Fig. 4(b) show that the "apparent" addition of turbulence due to rotational motion is higher than the inherent DWL averaging across all simulated atmospheres with values $\sigma_{U,\mathrm{FDWL}}/\sigma_{U,a}$ ranging from 1 to $\simeq 1.05$ on average. On the other hand, the effect of translational motion is only noticeable for low $\sigma_{U,a}$ values, where ratios higher than 1.3 are obtained. This is because the amplitude of the translational velocity experienced by the FDWL (0.25 m s^{-1}) is only comparable to wind velocity fluctuations of similar magnitude. The fixed turbulence boxes used by Peña et al. [35] were generated to match $\sigma_U \simeq 1.34$ m s^{-1} (dashed blue line), which is comparatively higher than the typical translational velocity values observed experimentally (< 1 m s^{-1} [20]), partially explaining the undercompensation observed under low turbulence levels in that latter study.

B. Campaign Overview

Fig. 5 presents a comparison between the 10-min mean HWS measured by the FDWL (U_{FDWL}) and the reference measurements of the metmast anemometers (U_a). The results are consistent with previous FDWL measurement campaigns [12], [16], [23], [35], showing an excellent agreement between the HWS values obtained from the FDWL and the metmast anemometers at all measurement heights.

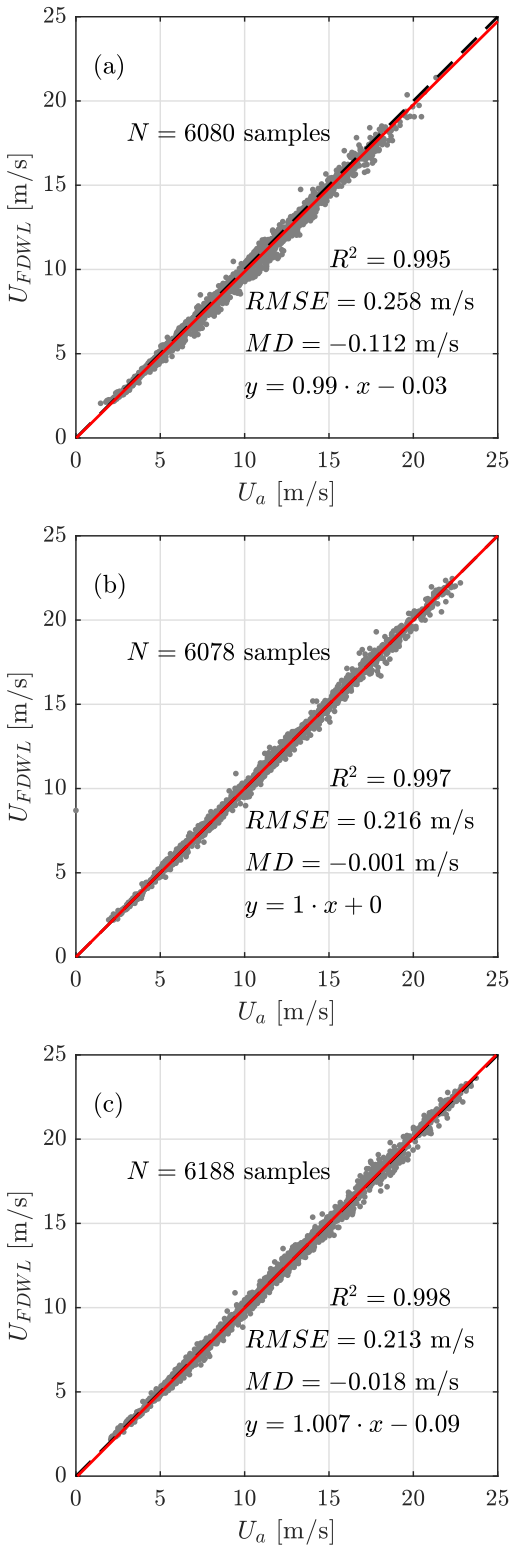


Fig. 5. Scatter plot comparing the 10-min mean HWS measured by the EOLOS[1] FDWL (U_{FDWL}) against the one measured by the IJmuiden metmast anemometers (U_a) at: (a) 25-, (b) 56-, and (c) 83-m height.

Fig. 6 shows the raw comparison, that is, without applied correction, between the standard deviation measured by FDWL HWS ($\sigma_{U,\mathrm{FDWL}}$) and the reference values measured by the metmast anemometers ($\sigma_{U,a}$) at the three measurement heights. Most of the records fall above the ideal 1:1 line, showing that, in most cases, the motion-induced "apparent" turbulence

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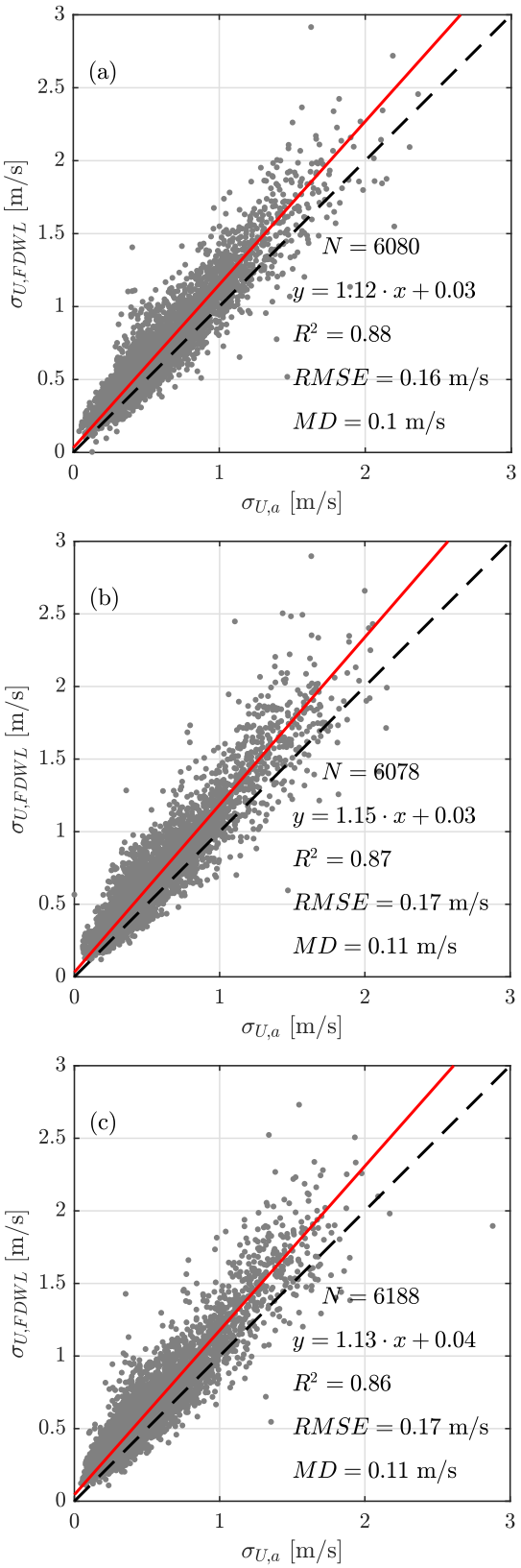


Fig. 6. Scatter plot comparing the 10-min HWS standard deviation measured by the EOLOS™ FDWL ($\sigma_{U,\mathrm{FDWL}}$) against the one measured by the IJmuiden metmast anemometers ($\sigma_{U,a}$) at the same height. Measurement heights (Section III): (a) 25, (b) 56, and (c) 83 m.

addition experienced by the FDWL is greater than the inherent averaging of the DWL [20], [53]. This is further evidenced

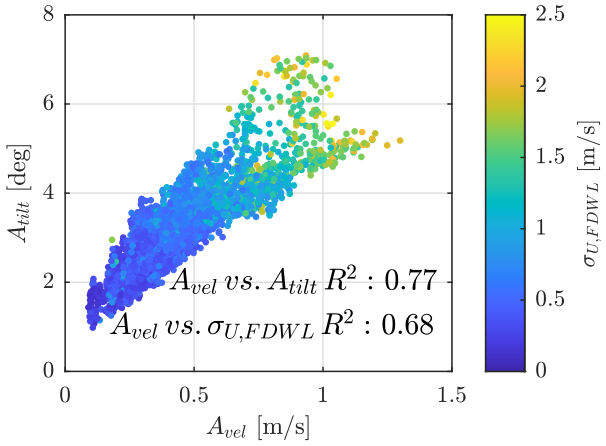


Fig. 7. Motion and turbulence experienced by the FDWL in the IJmuiden campaign. FDWL mean tilt amplitude [A_{tilt}, (9)] as a function of FDWL mean velocity norm [A_{vel}, (10)], and color-coded by the FDWL-measured HWS standard deviation ($\sigma_{U,\mathrm{FDWL}}$).

by linear-regression slopes > 1 as well as by positive MD values. This highlights the harsh wave conditions experienced by the FDWL during the IJmuiden campaign [27], [28]. Moreover, the higher the measurement height, the worse the correspondence. This is partly due to the larger scanning cone circle at higher altitudes. The coefficient of determination reduces from $R^2 = 0.88$ (25 m) to $R^2 = 0.86$ (83 m), and RMSE slightly increases from 0.16 m s^{-1} (25 m) to 0.17 m s^{-1} (83 m).

To quantify the rotational and translational motion experienced by the FDWL, we use the mean tilt amplitude (A_{tilt}) and the mean velocity norm (A_{vel}) variables, respectively [16]. The mean tilt amplitude is formulated as

$$A_{\mathrm{tilt}} = \sum_n \frac{1}{N} \sqrt{\mathrm{roll}_n{}^2 + \mathrm{pitch}_n{}^2} \tag{9}$$

where roll_n and pitch_n are the roll and pitch nth measurements by the IMU out of a total of N measurements in a 10-min period. The mean velocity norm is formulated as

$$A_{\mathrm{vel}} = \sum_n \frac{1}{N} ||\vec{t}_n|| \tag{10}$$

where $\vec{t}_n$ is the nth translational velocity vector measured by the IMU.

Fig. 7 explores the domain of motion and turbulence experienced by the FDWL in the IJmuiden campaign. It shows that the mean velocity norm of FDWL [A_{vel}, (10)] and the tilt amplitude [A_{tilt}, (9)] are well correlated ($R^2 = 0.77$). In addition, A_{vel} and $\sigma_{U,\mathrm{FDWL}}$ are also well correlated ($R^2 = 0.68$) because the excess of turbulence is partly due to the motion.

C. Motion-Correction Results: Basic and Enhanced Methods

The performance of the basic and enhanced motion correction methods at the 10-min level was analyzed by comparing the wind speed standard deviation measured by the FDWL, with and without correction, that is, $\hat{\sigma}^{\mathrm{corr}}_{U,\mathrm{FDWL}}$ and $\sigma_{U,\mathrm{FDWL}}$, respectively, against that of reference metmast anemometer

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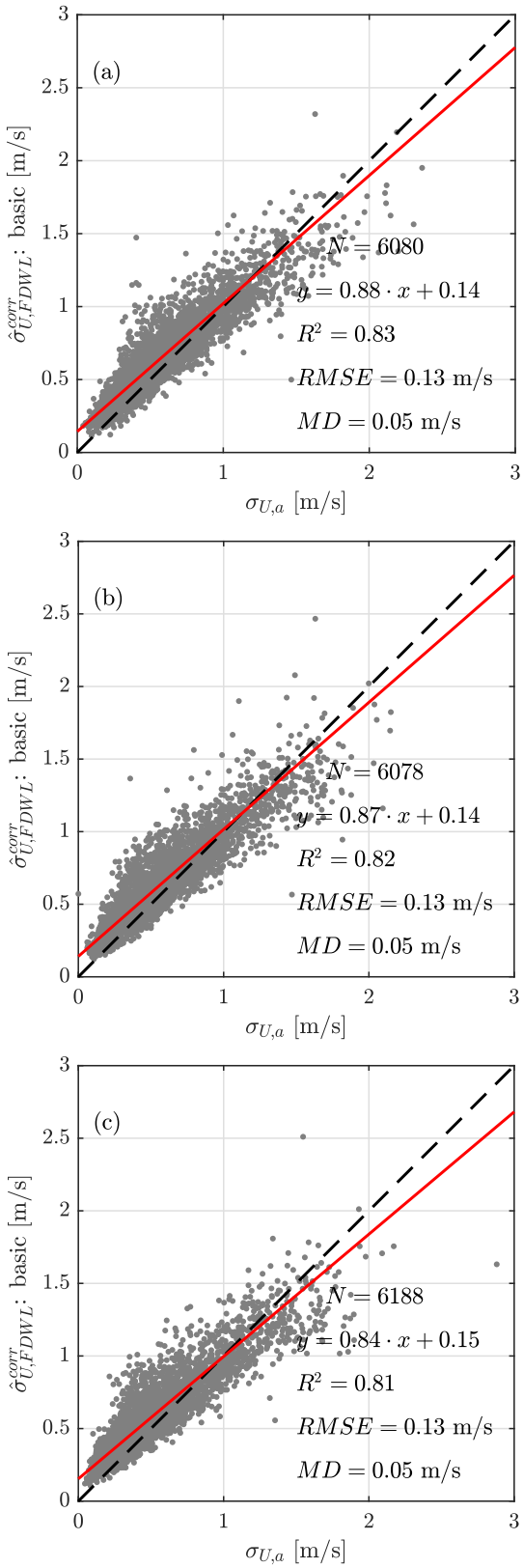


Fig. 8. Same as in Fig. 6 but with the FDWL records corrected with the *basic method* ($\sigma^{\text{corr.}}_{U,\text{FDWL}}$). Measurement heights (Section III): (a) 25 m, (b) 56 m, and (c) 83 m.

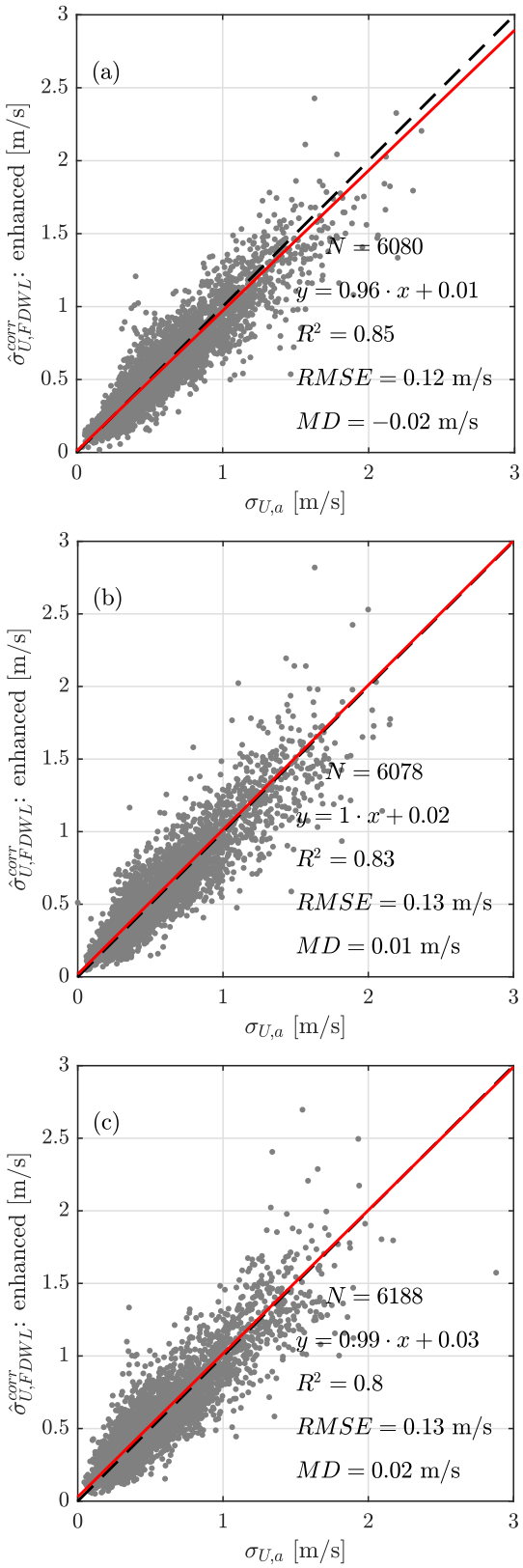


Fig. 9. Same as in Fig. 6 but with the FDWL records corrected with the *enhanced method* ($\sigma^{\text{corr.}}_{U,\text{FDWL}}$). Measurement heights (Section III): (a) 25 m, (b) 56 m, and (c) 83 m.

measurements $\sigma_{U,a}$. The characteristic parameters of the Mann turbulence boxes used for the correction methods are summarized in Table II.

Basic Method Fig. 8 shows the motion-correction results using the basic method. The scatter is aligned more closely with the ideal 1:1 line when compared to the results without

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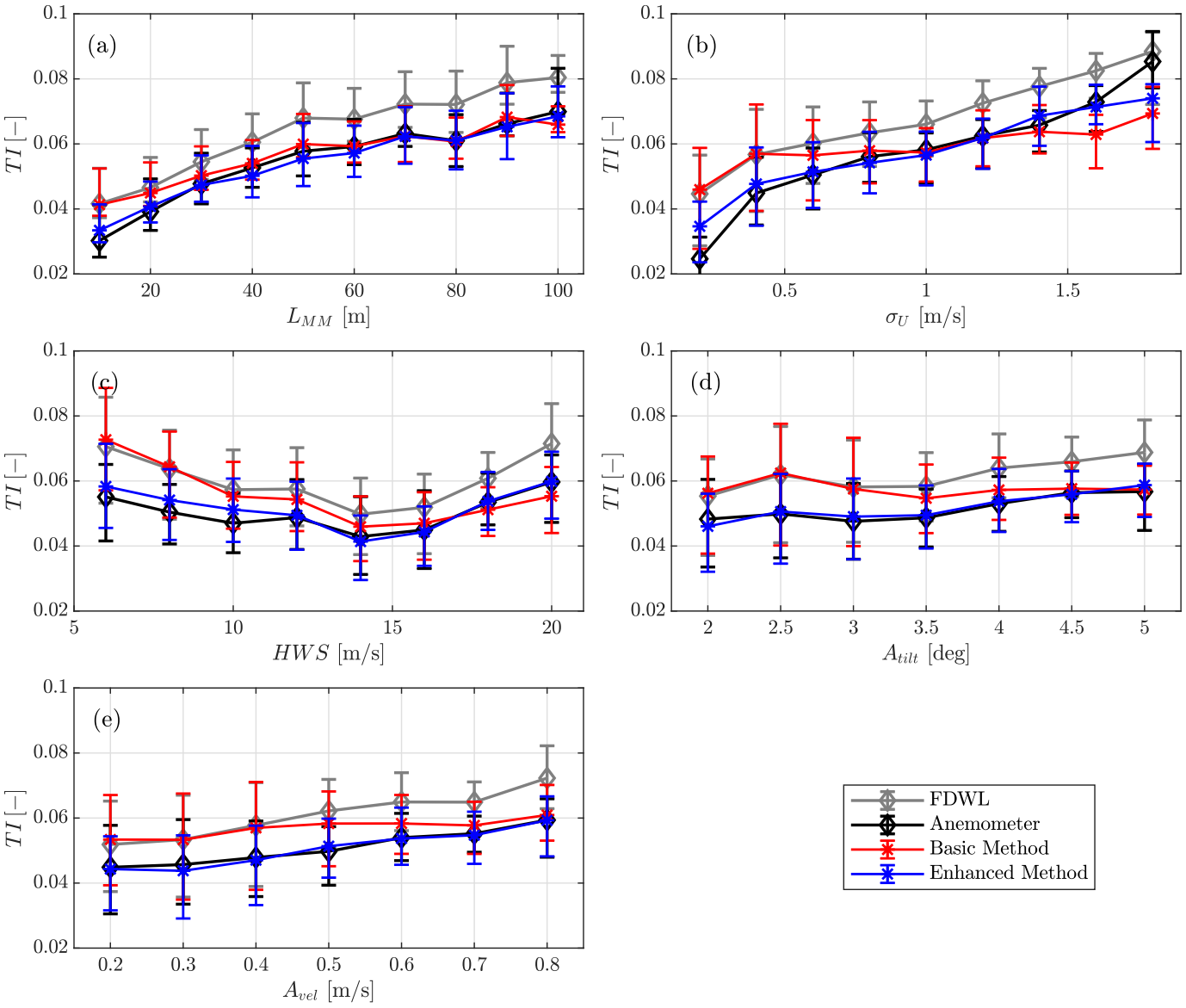


Fig. 10. FDWL compensation performance as a function of different atmospheric and motion conditions clustered by (a) L_{MM}, (b) σ_U, (c) U, (d) A_{tilt}, and (e) A_{vel}. Gray and black traces represent the uncorrected FDWL and reference anemometer TI measurements at 83-m height, respectively. Markers illustrate cluster medians. Lower and upper bars represent the 25th and 75th percentiles of each cluster.

TABLE II
TURBULENCE-BOXES PARAMETERS USED FOR THE MOTION-CORRECTION BASIC AND ENHANCED METHODS

Parameter	Basic Method	Enhanced Method
$\alpha\epsilon^{2/3}$	0.05	0.05
L_{MM}	61	From 0 to 70 in steps of 2
Γ	3.2	3.2
σ_U	1.34	From 0 to 2.5 in steps of 0.3
L_x, L_y, L_z	18000, 128, 128 m	
n_x, n_y, n_z	4096, 32, 32 points	
Seeds	From -1 to -21 in steps of -3	

compensation (see Fig. 6). The statistical indicators evidence this correction, being MD reduced from MD $\simeq$ 0.1 m s^{-1} (without compensation, Fig. 6) to MD $=$ 0.05 m s^{-1}, and RMSE decreasing from RMSE $\simeq$ 0.17 to $\simeq$ 0.13 m s^{-1}. Linear regression lines highlight *undercompensation* for $\sigma_{U,\mathrm{FDWL}} <$ 1 m s^{-1}. This is likely due to the use of turbulence boxes with fixed Mann parameters, which portray an atmosphere with higher turbulence [$\sigma_U \approx$ 1.34 m s^{-1}, vertical dashed blue line in Fig. 4(b)] than the measurements. The effect of translational motion on the measurement error is, therefore, greater in the observations than what can be simulated, as the real motion amplitude (A_{vel}) is in reality comparable to the measured σ_U. Most translational motion observed under such scenarios ($A_{\mathrm{vel}} <$ 0.5 m s^{-1}; see Fig. 7) is not adequately captured by the basic method. For example, at 0.25 m s^{-1} surge and $\sigma_{U,a} =$ 0.2 m s^{-1}, the necessary motion compensation correction factor is $\simeq$ 0.83, whereas the basic method accounts for a factor 1.1. Moreover, since the basic method assumes a fixed-length scale, the DWL inherent turbulence averaging is underestimated [see Fig. 4(a)] and, at the same time, counteracted by the rotational motion variance addition, leading to a final correction factor of $\simeq$ 1.

Enhanced method Fig. 9 illustrates the motion-correction results using the enhanced method. The superior performance

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of the enhanced method with respect to the basic becomes evident, with the scatter lying symmetrically along the 1:1 line. Regression lines exhibit slope values ranging from 0.98 to 1.00, and MDs, being reduced from MD $\simeq$ 0.05 to $\simeq$ 0.01 m s^{-1}. Nevertheless, the RMSE showed similar values, and the scatter remains higher than in the case without any compensation. This can be attributed, first, to atmospheric stability regimes deviating from the neutral condition assumed in the model, which slightly alters the proportionality constant between σ_U and u_* used in 7 [54]. Second, small random errors introduced by the 2P-algorithm and the Högström polynomial profile [19], [55] may also contribute to the observed scatter.

The proposed enhanced method outperforms all previously tested state-of-the-art approaches used on data from the same campaign—such as the unscented Kalman filter [33] and the statistical error correction method [15]—both of which fail to correct for the inherent averaging of DWLs (|MD| $>$ 0.05, RMSE $>$ 0.15 m/s, and R^2 $<$ 0.7 when comparing the corrected FDWL against the reference sonic anemometers). However, this improved performance comes at the cost of significantly higher computational demands (see Section III-E). As a result, the method is currently suitable only for postprocessing, whereas the unscented Kalman filter can be executed directly on the FDWL's onboard CPU due to its low memory and processing requirements. Despite these limitations, the enhanced method remains the only known approach capable of simultaneously compensating for both FDWL motion and inherent averaging.

D. TI Clustering Analysis

To complete the analysis, the correction accuracy of the basic and enhanced methods was evaluated in terms of TI under various motional, wind speed, and turbulence conditions. The clustering variables used in this analysis were: L_{MM}, σ_U, U, A_{tilt}, and A_{vel}. Clusters containing fewer than 100 samples were considered nonrepresentative and excluded from the analysis.

Fig. 10 presents the TI clustering results. The FDWL-measured TI (gray trace) appears consistently higher than the anemometer reference (black trace) across all atmospheric and motion scenarios, indicating greater apparent turbulence addition due to motion than reduction due to the scanning-cone inherent averaging.

The *basic method* (red trace) achieves accurate correction under atmospheric scenarios that approximately correspond to those portrayed by the turbulence boxes with fix parameters ($\sigma_U \approx 1.34\ m \cdot s^{-1}$ and $L_{\mathrm{MM}} = 61$ m), that is, scenarios in which we estimate values of $L_{\mathrm{MM}} > 40$ m and σ_U between 1 and 1.5 m s^{-1} in Fig. 10. Outside these ranges, as expected, the basic method tends to undercompensate the TI (panel b). Negligible compensation can be observed in Fig. 10(d) for $A_{\mathrm{tilt}} < 4.5°$ and Fig. 10(e) for $A_{vel} < 0.6$ m s^{-1} (note also that A_{tilt} and A_{vel} are variables highly correlated to σ_U; refer to Fig. 7).

In contrast, overcompensation is found in highly turbulent scenarios (σ_U $>$1.5 m s^{-1}), in which the simulated turbulence is lower than the observed, and thus, the translational motion-induced error is over-represented. On the other hand, the *enhanced method* (Fig. 10, all panels, blue trace) shows an overall improvement of the results obtained by the basic method. It correctly compensates TI in all atmospheric and motion scenarios, with TI values accurately representing those of the anemometer in all clusters. Some undercompensation is still observed in low-turbulence (σ_U $<$ 0.5 m/s) and low-wind-speed (HWS $<$ 10 m/s) conditions. We hypothesize that this results from a slight overestimation bias in the friction velocity (u_*) reported for the 2P algorithm in marine environments [55]. This bias (approximately 0.075 m/s) leads to an a priori overestimation of σ_U by about 0.2 m/s [see (7)], which, in turn, causes an underestimation of the translational motion-induced error in low-turbulence scenarios [see Fig. 4(b)].

V. Summary and Conclusion

An *enhanced method* to obtain anemometer-like measurements of turbulence from FDWLs was presented. The method is based on the basic correction technique presented by Peña et al. [35], which uses turbulence boxes generated by the Mann model to simulate FDWL and anemometer measurements under turbulent wind fields. In contrast to the *basic method,* which uses turbulence boxes with fixed turbulence characteristics, the novelty of the enhanced method is the use of turbulence boxes that closely match the atmospheric turbulence characteristics in which the FDWL measures.

The Mann model parameters for the simulations are indirectly estimated from measurements of the FDWL. More precisely, we estimate the friction velocity and wind shear gradient via the so-called 2P algorithm to compute the turbulence length scale L_{MM}. The other two Mann parameters are assumed constant; however, the turbulence boxes are scaled to match the velocity variances, which are computed from their relation with the friction velocity according to MOST [(7)].

A parametric study highlighted the *error factors affecting FDWL measurements* of turbulence (Fig. 4): FDWLs were found to be greatly affected by translational motion when its amplitude is comparable to σ_U. Moreover, the impact of the DWL's scanning-cone inherent averaging effect was found to be larger when turbulent length scales (L_{MM}) were small in comparison to the cone diameter. The basic method underestimates these effects as it uses turbulence boxes with fixed Mann parameters.

The *basic and enhanced methods* were *validated* with measurements from the IJmuiden campaign, by comparing an FDWL against reference anemometers on a metmast for 3 months. The *mean* HWS measured by FDWLs was virtually identical to the anemometer's reference [Fig. (5)]. Regarding the motion-corrected HWS standard deviation, the basic method showed a fair degree of correction (Fig. 8) in contrast to the uncorrected case (Fig. 6), improving statistical indicators from $RMSE \approx 0.17$ to ≈ 0.13 m s^{-1}, and MD ≈ 0.11 to ≈ 0.05 m s^{-1}. However, undercompensation of turbulence was found for σ_U $<$ 1 m s^{-1}, due to underestimation of the translational motion contribution to the error by the basic method. The enhanced method properly corrects turbulence under such turbulence scenarios, achieving indicators as good as RMSE ≈ 0.12 m s^{-1} and |MD| ≈ 0.02 m s^{-1}.

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The basic and enhanced methods *motion-correction performance* were also analyzed in terms of TI through a cluster analysis (Fig. 10). The basic method was only able to provide accurate correction when the observed turbulence conditions exhibited parameters similar to those from the turbulence boxes generated with fixed parameters in [35], that is, $L_{\mathrm{MM}} = 61$ m and $\sigma_U \approx 1.34$ m s^{-1}.

All in all, this study demonstrates the importance of accurately modeling the observed turbulence characteristics for the successful motion compensation of FDWL measurements using turbulence boxes. By considering this, measurements of the HWS and TI close to those of metmast anemometers can be achieved from FDWL measurements.

Further research should aim to extend the results presented here to alternative measurement campaigns and instrumental configurations. The enhanced method can be adapted for different FDWL instruments by updating the FDWL simulator geometry and/or measurement procedures. Although generally applicable to the surface layer of open-sea locations, the empirical relationships employed by the enhanced method [(8) and (7)] should be validated through additional experimental campaigns. Moreover, the enhanced method currently assumes a neutral atmosphere; this can be further extended to account for atmospheric stability variations. Additionally, the effects of systematic and random errors from the 2P algorithm and the Högström polynomial model on the enhanced method need to be quantified. Lastly, accurately simulating the FDWL probe length could further improve the correction results.

APPENDIX

A. Friction Velocity and Wind-Shear Retrieval

The two-parameter-retrieval algorithm (the 2P algorithm for short) by Araújo da Silva et al. [12] combined with the Högström wind profile polynomial model [56] can provide an estimate of both the friction velocity u_* and the wind local gradient dU/dz from the FDWL-measured HWS vertical profile, respectively.

2P algorithm Araújo da Silva et al. [12] presented a parametric solver algorithm to estimate atmospheric stability and friction velocity from vertical profiles of HWS. Based on the MOST [45], a fit to the logarithmic wind profile model described by MOST to experimental wind profile measurements is performed. Over the sea, and taking advantage of the Charnock relation for the roughness length, the logarithmic wind profile can be well described by two constitutive parameters, the Obukhov length (L) and friction velocity (u_*). The 2P algorithm estimates these two parameters by optimizing them to fit the MOST model to the measured wind profile. Formally

$$\left(\hat{L}, \hat{u}_*\right) = \arg\min_{L,u_*} ||\boldsymbol{U}_{\mathrm{obs}} - \boldsymbol{U}_{\mathrm{MOST}}(L, u_*)||^2 \tag{11}$$

where $\boldsymbol{U}_{\mathrm{obs}}$ is the observed wind profile and $\boldsymbol{U}_{\mathrm{MOST}}$ is the logarithmic wind profile model from MOST.

Multiple studies have proved the validity of friction velocity estimates using the 2P algorithm [12], [18], [55], with coefficients of determination R^2 greater than 0.8 and RMSE values as low as 0.08 m s^{-1} when using sonic anemometers as reference.

Högström polynomial profile. The wind profile polynomial proposed by Högström [56]

$$U(z) = U_0 + A\ \ln(z) + B\ln(z)^2 \tag{12}$$

where U_0, A, and B are parameters obtained by fitting the polynomial to an observed wind profile, similar to the 2P algorithm in (11), describes well the behavior of the wind speed in the lower atmosphere. Also, the wind local gradient can be readily estimated from the derivative of (12)

$$\frac{d\hat{U}}{dz} = \frac{A + 2B\ \ln(z)}{z}. \tag{13}$$

B. Measurement Simulations in Turbulence Boxes

1) Turbulence Boxes Setup: We define the right-handed Cartesian XYZ inertial frame of reference. The FDWL lens, that is, the origin of the lidar deflected light beam, is "fixed" at the (0,0,0) m point of the XYZ frame of reference [see Fig. 11(a)]. The turbulence boxes are centered at $(x, y, z) = (x, 0, h)$, where h is the lidar measurement height and the free variable x denotes that the boxes are moved along the x-axis with a constant speed following Taylor's frozen atmosphere hypothesis [57].

2) Sisec: FDWL Simulation Conditions: Next, we summarize the procedure of the "FDWL *simulator*" block of Fig. 2. The 6-DoF geometry used is from [20] and is derived from [16]. The FDWL IMU measures the buoy's 6-DoF motion in a North-East-Down (NED) global frame of reference. Therefore, for simulations, we align the N, E, and D IMU axes with the X-, Y-, and Z-axes of the XYZ inertial frame of reference, respectively [see Fig. 11(b)].

N wind-vector retrievals are simulated within a turbulence box simulation [note that north (N) is the direction from which the wind blows in Fig. 11(a)]. The ZephIR^{TM} 300 takes 50 radial velocity measurements for each scan of the instantaneous wind vector (see Section II). Thus, for the ith scan at time t_i, we need to simulate the $k = 0, \ldots, 49$ radial velocities $u_{r,k}$ along the scanning cone. We depart from a motionless scenario, in which the 50 lidar radial velocity vectors in a scan ($\hat{r}_k$) can be well described within the scanning cone aperture angle [$\theta_0 = 30°$, Fig. 11(b)] by 50 uniformly spaced azimuth angles (ϕ_k, with $k = 0, \ldots, 49$) starting from the initial azimuth scanning phase ϕ_0. The radial velocity vectors are then rotated according to the IMU-measured tilt [Ω_r, Ω_p, and Ω_y, for roll pitch and yaw, respectively, in Fig. 11(b)] through Euler's rotation matrices [58], obtaining the 50 "true" radial velocity vectors $\hat{r}_k^{\mathrm{rot}}$ in a scan (deep blue vectors in Fig. 11). Thus, if the IMU sampling times do not match the ones of the lidar, they need to be resampled. The lidar radial position vector $\vec{r}_k$ (gray vectors in Fig. 11) can be obtained by multiplying the focus distance d to the rotated lidar radial unitary vectors ($\hat{r}_k^{\mathrm{rot}}$) as

$$\vec{r}_k = d \cdot \hat{r}_k^{\mathrm{rot}} \tag{14}$$

with the focus distance being $d = h/\cos(\theta_0)$. Then, we retrieve the box's 50 wind vectors $\vec{u}_k^{LoS}$ (red vectors in Fig. 11) closest to the 50 lidar radial position vectors $\vec{r}_k$. The 50 radial velocity measurements $u_{r,k}$ are computed as the dot

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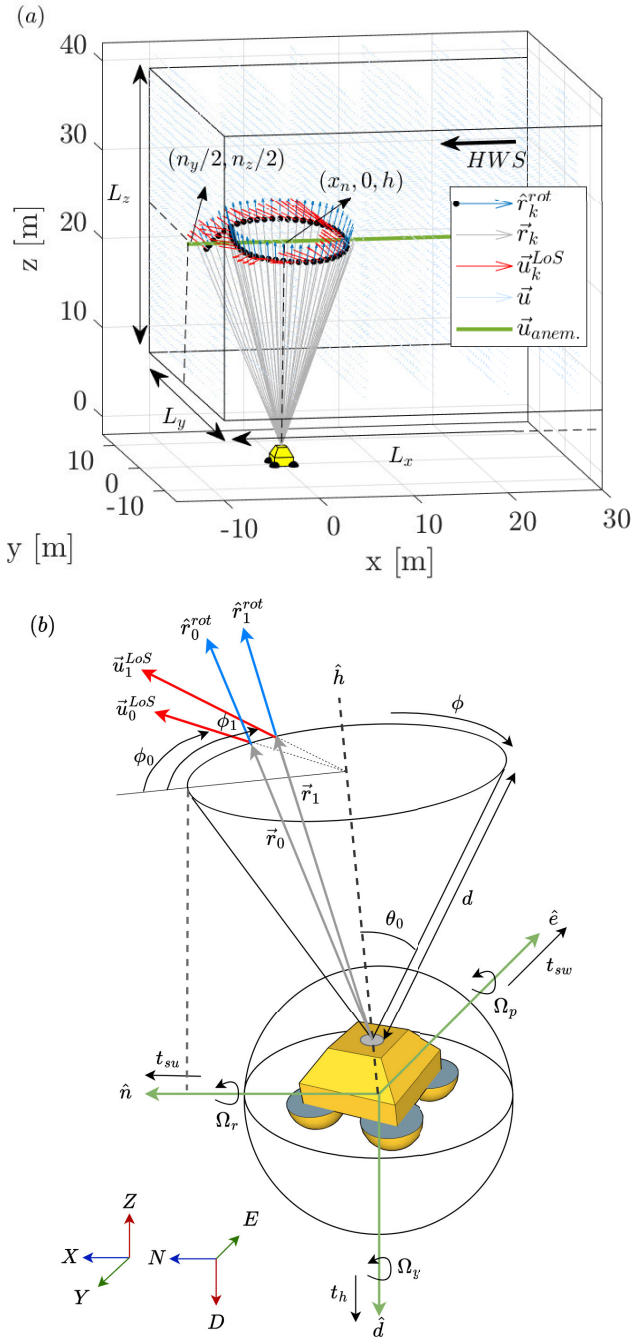


Fig. 11. Schematic of the turbulence boxes simulation geometry. (a) Schematic of the FDWL and anemometer measurement simulation setup within the turbulence boxes. (b) Detail of the FDWL measurement geometry and motion.

product between the rotated lidar radial unitary vectors ($\vec{r}_k^{\text{rot}}$) and their correspondent wind vector ($\vec{u}_k^{\text{LoS}}$) as

$$u_{r,k} = \left(\vec{u}_k^{\text{LoS}} - \vec{t}\right) \cdot \hat{r}_k^{\text{rot}} \tag{15}$$

with $\vec{t}$ being the FDWL translational velocity vector ($\vec{t} = [t_{\text{su}}, t_{\text{sw}}, t_h]$). Finally, the wind vector measured by FDWL $\boldsymbol{u}_n^{\text{FDWL}}$ at time t_n is recovered using the VAD algorithm [9] from the 50 radial velocities (see Section II-A). From the N FDWL-retrieved wind vectors in a turbulence box, we can compute the *corrupted* HWS mean (U_{FDWL}), HWS standard deviation $\sigma_{U,\text{FDWL}}$, and TI products according to the expressions in Section III-A.

3) Simulation of Anemometer Measurements in Turbulent Conditions: Finally, we define the procedure of *"Anemometer Simulator"* block of Fig. 2. To simulate *reference* anemometer measurements within a box, the anemometer is "fixed" at the lidar measurement height (0,0,h), and the turbulence boxes are again centered at $(x, y, z) = (x, 0, h)$. Therefore, anemometer wind vector measurements $\boldsymbol{u}_m^{\text{anem}}$ are simply the set of wind vectors along the x-direction of the turbulence box at height h [green line in Fig. 11 (a)]. Similarly, from the wind vectors retrieved from the anemometer, we can compute the *reference* 10-min products following the expressions in Section III-A.

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